
AI Code Review & Security Analysis Agent

Milestone 1 Completion Report

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1. Project Overview

The AI Code Review & Security Analysis Agent is a robust, AI-powered tool designed for deep code analysis, real-time feedback, and security vulnerability detection. Milestone 1 successfully establishes the foundational architecture, the Retrieval-Augmented Generation (RAG) pipeline, syntax validation services, and a professional, interactive client interface.

2. Completed Modules for Milestone 1

2.1 Client Interface (Frontend)

- Developed a responsive Single Page Application using React and Vite.
- Integrated Monaco Editor for a VS Code-like coding experience (Python & Java support).
- Implemented a premium, state-of-the-art Dark Mode UI featuring glassmorphism and micro-animations.
- Built interactive components for Security Advice Cards and collapsible Error Snippets.

2.2 Code Submission & Validation Module (FastAPI Backend)

- Built RESTful API endpoints for code submission (via direct paste or file upload).
- Integrated Python 'ast' and Java 'javalang' libraries for strict syntax and structural validation.
- Configured a PostgreSQL database using SQLAlchemy to persist scan history and results.

2.3 Security Analysis & RAG Pipeline

- Set up the PostgreSQL vector database utilizing the 'pgvector' extension.
- Created ingestion scripts to process and chunk Markdown-based knowledge base files (e.g., OWASP, Secure Coding).
- Integrated HuggingFace's 'all-MiniLM-L6-v2' model to generate embeddings for semantic search.
- Developed retrieval logic to fetch relevant security context based on validated code submissions.

2.4 Infrastructure & Containerization

- Created Docker and Docker Compose configurations to orchestrate the FastAPI backend and PostgreSQL database.
- Ensured seamless local development setup and environment consistency.

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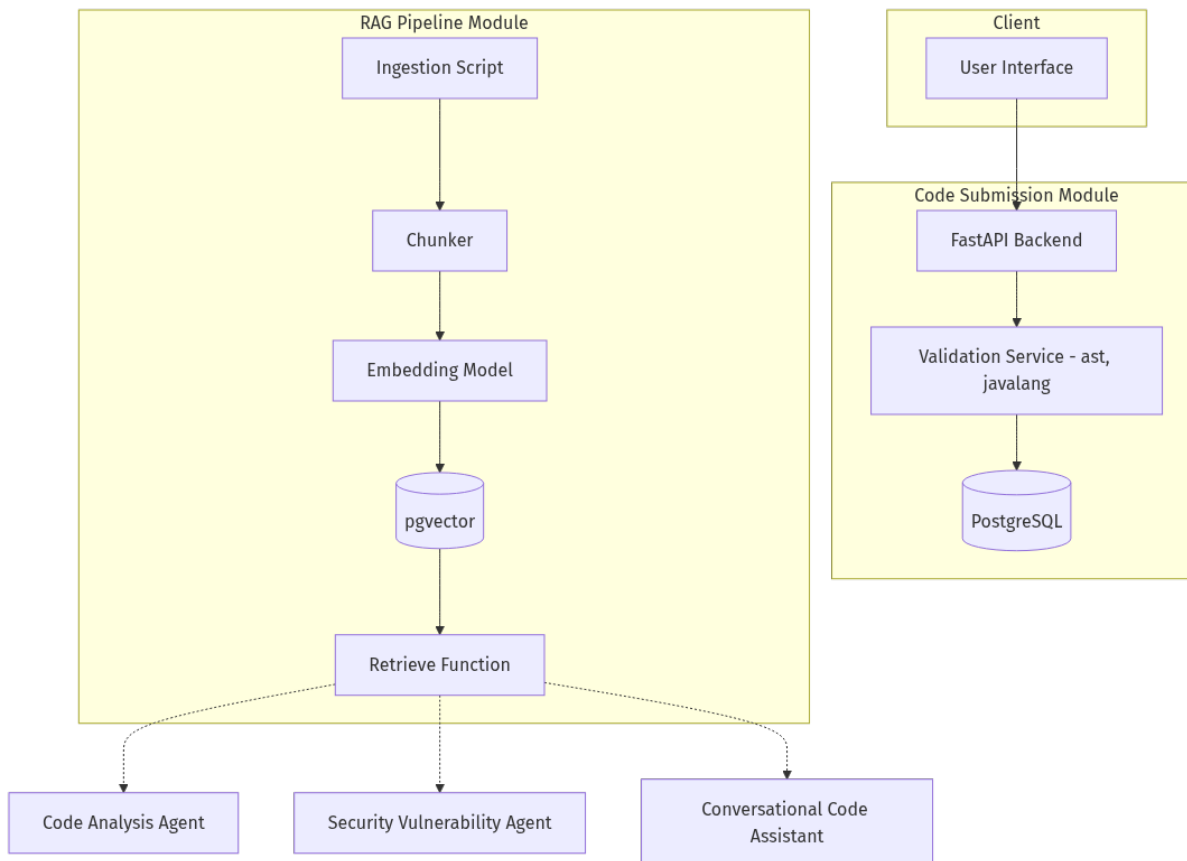
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3. System Architecture Diagram

The following diagram outlines the high-level architecture of the system:



4. Source Code Repository

The complete source code and documentation for Milestone 1 are available at:

<https://github.com/Mugunthan-127/AI-Code-Review-Security-Analysis-Agent>

5. Next Steps (Milestone 2+ Roadmap)

- Remediation Agent: Automatic generation of code fixes and security patches.
- Conversational Assistant: Interactive chat to seamlessly explain vulnerabilities.
- PR Summary Agent: Automated generation of pull request review summaries for CI/CD integration.